Predictive Physical Realizability: A Unified Planning–Execution Feedback Architecture for Motion Planning

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Abstract

Motion planners — sampling-based, optimization-based, or fixed-structure time-domain planners — certify geometric and kinematic feasibility: that a path is collision-free and respects declared velocity/acceleration bounds. None of the widely deployed planning stacks certify *dynamic physical realizability*: whether the robot's actuators, given its current configuration, contact state, payload, and disturbance environment, can actually execute the planned motion as time progresses. Because actuator authority is state- and environment-dependent, a trajectory can be kinematically feasible and still drive one or more joints toward saturation in the near future. This paper introduces **predictive physical realizability**, a feedback signal that predicts future loss of actuator authority over a receding horizon and feeds it back into motion planning before the loss becomes an execution failure. We define a horizon-wide realizability certificate conditioned on predicted state, contact, payload, and disturbance information; a hierarchical planner response — retime, reshape, reroute, or safely brake — governed by whether local adaptation can restore feasibility along the current route; and conditions under which a route-level change is provably necessary. The architecture preserves the fixed-structure computational philosophy of the underlying local planner for certificate evaluation and Level 0–1 (retiming): state- and environment-dependent information enters through linear/vector terms rather than rebuilding the optimization structure online. This property does not currently extend to Level 2 (reshaping), whose QP is rebuilt from scratch each call and measured at a real, disclosed cost (§IX). We position the contribution against reference-governor theory, actuation-aware legged planning, and MoveIt 2's Hybrid Planning architecture, and propose a benchmark suite — reachable today on a fixed-base manipulator, with legged/humanoid extensions explicitly scoped as future work pending a resourcing decision — designed to isolate what predictive feedback adds over reactive saturation handling.

I. Introduction

Robot motion planning is conventionally organized as a pipeline: a global or sampling-based planner searches for a geometrically valid path, a time-parameterization stage converts that path into a timed trajectory subject to declared velocity/acceleration bounds, and an execution layer tracks the result. This separation is effective when the robot's dynamic capability is state-independent, but it is not: the torque required to realize a given acceleration depends on configuration, velocity, active contacts, payload, interaction forces, and external disturbance, through

$$\tau = M(q)\,u + C(q,\dot{q})\,\dot{q} + g(q) + J_c^\top F_c + \tau_{\text{ext}}.$$

A trajectory that is geometrically and kinematically valid at planning time can therefore still drive an actuator toward saturation once contact, payload, or disturbance conditions change — a fact that current pipelines discover only when it happens, not before. We write this as

$$\text{kinematically feasible} \;\not\Rightarrow\; \text{physically realizable}.$$

MoveIt 2 as the concrete anchor for this gap. We do not treat this as a hypothetical failure mode. MoveIt 2's default pipeline — OMPL for geometric search, Time-Optimal Trajectory Generation (TOTG) as the default time-parameterization stage, with optional Ruckig jerk-limited smoothing — is exactly the path-first architecture this paper is positioned against. One directly verifiable, illustrative data point is documented in moveit/moveit2#2600 [1]: TOTG reconstructs an acceleration of roughly 200 rad/s² at a path inflection point against a declared limit of 0.2 rad/s². We cite this one issue because it is directly verifiable and reproducible, not because it establishes a general pathology in TOTG or MoveIt 2 as a whole — the issue's own reporter traces the specific failure to an unsupported edge case (a full 180° joint turn) and a default `path_tolerance=0.0` in the raw TOTG class that MoveIt's standard `RobotTrajectory` wrapper already avoids by setting a different default, and we state that qualification here rather than let the citation imply more than the source supports (see *Draft Status and Open Items*, item 1). Two additional reports referenced in an earlier internal draft of this work could not be re-located on a follow-up search and are *not* cited here. Independently of any specific bug, MoveIt 2's own **Hybrid Planning** architecture already separates a slower global planner from a faster, recurrent local planner and explicitly supports event-driven, sensor-reactive replanning — a real, currently-unoccupied slot for exactly the kind of feedback this paper proposes, not a contrived integration target.

Positioning relative to OMPL. OMPL answers the question *does a collision-free path from start to goal exist?* This paper asks the strictly downstream question: *does a physically realizable trajectory exist along this path, given the robot's configuration, contact state, payload, and predicted environment?* OMPL is not a competitor at any point in this paper; it is the standard upstream global planner, and the contribution sits at the local-planner slot of MoveIt's own Hybrid Planning architecture, consuming OMPL's output rather than replacing it.

We do not claim that conventional planning pipelines are useless. We claim that their feasibility representation is incomplete, because actuator authority is a function of state and environment that the geometric/kinematic feasibility check does not see.

Key research question. *Can future loss of physical realizability be predicted during execution and fed back to motion planning early enough that the robot can adapt, reroute, or safely brake before insufficient actuator margin causes task failure, collision, or falling?* (The certificate predicts $m_{\text{phys}} < m_{\text{safe}}$ — a margin threshold, deliberately set to trigger well before actuator saturation — not "loss of control" in a stronger, dynamical-systems sense; this paper does not claim the latter and uses "actuator margin/authority" throughout to keep the distinction visible.)

A. Contributions

1. **A problem formulation** that distinguishes geometric/kinematic feasibility from future actuator realizability, and defines the combined safe set $\mathcal{F}_{\text{safe}} = \mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}} \cap \mathcal{F}_{\text{dyn}}$ that existing pipelines do not jointly certify.
2. **A predictive realizability certificate** m_{phys}, conditioned on predicted state, contact, payload, and disturbance information, that bounds actuator margin over a receding horizon without requiring the local planner's optimization structure to be rebuilt online.
3. **A hierarchical planning response** — execute, retime, reshape, reroute, or safely fall back — together with a checkable, sufficient condition (Theorem 3) for when the architecture's own implemented local-adaptation mechanism has been exhausted and route-level replanning is warranted, distinct from the (true but definitional) idealized statement that no modification of an infeasible route can ever be feasible.
4. **A real-time architecture and benchmark design** targeting MoveIt 2's Hybrid Planning interface, with an experiment suite that isolates the contribution of prediction, adaptation, and rerouting individually via ablation.

We deliberately do *not* claim to be the first work to consider actuator limits in motion planning, to have invented margin-based feedback, or to guarantee safety under arbitrary disturbance — see §XI and *What This Paper Does Not Claim* below. The novelty we claim is narrower: a predictive feedback interface that converts future physical-realizability information from the execution layer into planner-level adaptation and route-level replanning, unified across geometric and physical failure modes under one hierarchical response.

II. Related Work

A. Motion planning and time parameterization. Sampling-based planners such as OMPL certify geometric feasibility; time-parameterization methods (TOTG, TOPP-RA, Ruckig) subsequently fit a time law respecting declared kinematic bounds. These methods do not represent configuration-, contact-, or payload-dependent actuator authority as a planning-time signal. A documented MoveIt 2 issue, moveit/moveit2#2600 [1], illustrates that nominal time-parameterization limits do not by themselves guarantee the reconstructed trajectory satisfies the intended acceleration bounds — TOTG there reconstructs an acceleration roughly 1000× the declared limit at a path inflection point — though its own reporter traces the specific failure to an unsupported edge case (a full 180° joint turn) and a default `path_tolerance=0.0` that MoveIt's standard `RobotTrajectory` wrapper already avoids by setting a different default (see *Draft Status and Open Items*, item 1); we do not present this as a general architectural pathology, only as one directly verifiable, illustrative data point.

B. Reactive and hybrid motion planning. MoveIt 2's Hybrid Planning architecture separates a slower global planner from a faster recurrent local planner and explicitly supports event-driven, sensor-reactive local adaptation, including to changing surface conditions. This paper's contribution is not the hybrid architecture itself, but a new feedback signal and local planning constraint that occupies that architecture's local-planner slot.

C. Reference governors and safety filters. Reference Governors and Explicit Reference Governors (Garone, Di Cairano, and Kolmanovsky, *Automatica* survey, 2017 [2]; origins in Bemporad, Casavola, and Mosca, 1997 [3]) predict a scalar margin to constraint violation — the Dynamic Safety Margin — and modulate a reference command to keep the closed loop inside a safe invariant set before violation occurs. This is conceptually the closest prior architecture to the margin-feedback idea at the center of this paper, and we do not claim novelty for "predict a margin and feed it back before a constraint is violated" as a general principle: that principle is three decades old. What is not addressed by the reference-governor literature, to our knowledge, is a margin specifically defined over actuator/torque authority, conditioned jointly on predicted contact, payload, and terrain information, coupled to a planner response that includes route-level replanning rather than only reference modulation within a fixed route.

A closely related recent architecture is the **Path Feasibility Governor** (PathFG; Zhang, Liu, and Liao-McPherson, arXiv:2507.09134, 2025 [4], read in full — see *Draft Status and Open Items*, item 4), which places a governor between a path planner and a nonlinear MPC controller: given a path $p(s)$, $s \in [0, 1]$, PathFG selects an auxiliary reference s_k along the path — as far toward the target as a constructed invariant terminal set permits — using the *previous* cycle's optimal terminal predicted state as a cheap, one-dimensional proxy for the true (intractable) N-step backward reachable set, so the update is a bisection search over $s \in [0, 1]$ rather than an online set computation. The high-level architecture is close to the one proposed here (§III–§V): both interpose a feasibility-filtering layer between a path/route planner and a lower-level controller. We differ in three ways. (i) The present work conditions the realizability certificate explicitly on predicted environment, contact, and payload information rather than the nominal path alone; PathFG's own text scopes disturbances out explicitly ("we focus on nominal MPC without explicitly considering disturbances for brevity"), deferring them to future pairing with a robust-MPC formulation, and has no payload, contact, or torque-margin concept in either its general theory or its worked quadrotor example. (ii) The present work defines an explicit taxonomy of geometric versus physical failure with a hierarchical, mechanism-differentiated response (§V); PathFG applies one uniform mechanism (auxiliary-reference selection) rather than a graduated retime/reshape/reroute/fallback response. (iii) The present work targets actuator/torque realizability specifically; PathFG's constraint sets $\mathcal{X}, \mathcal{U}$ are abstract polyhedra — thrust/rate bounds in its quadrotor example, but nothing in the general framework is torque-specific.

Two further points cut the other way and are stated here rather than left for a reviewer to find. First, PathFG proves what this paper's Theorem 1 explicitly does not: a *closed-loop* guarantee. Its Theorem 1 (recursive feasibility, by induction over invariant-set membership) and Theorem 2 (asymptotic stability, via an ISS lemma on the tracking error plus a finite-time-convergence argument bounding how long the auxiliary reference can stall) are complete, invariant-set-based proofs for the closed loop under the combined governor+MPC policy. This draft's own Theorem 1 is definitional over the *predicted* trajectory only and says so plainly ("it does not, by itself, say anything about closed-loop behavior once feedback, replanning, or an inaccurate $\Delta\tau_i$ are in play") — an honest asymmetry, not a minor stylistic gap, and one this paper does not yet close. Second, that rigor is bought by a real cost: PathFG's constructive terminal-set procedure (Riccati-based LQR terminal region, its Assumptions 4–6 and Lemma 3) is specific to linear or linearized systems, and its numerical validation (a quadrotor outer loop) is exactly such a linearized system — a natural fit for smooth continuous dynamics, but not a solved problem for a contact-mode-switching, actuator-margin-driven system where the feasible set's character changes across modes, which is closer to the regime this paper targets. This is offered as a reason the two papers take different technical routes (invariant sets vs. a direct per-step margin with an explicit uncertainty bound), not as a weakness unique to PathFG: both papers state their theory for a general nonlinear system and validate on a simplified instantiation of it (PathFG's linearized quadrotor; this draft's reduced-order planar arm, §IX), which is standard practice in this literature and worth naming as a shared limitation rather than implying it is peculiar to the present work's reduced-order study.

D. Actuation-aware and dynamics-aware planning. Prior work has incorporated torque limits, feasible wrench sets, and dynamic feasibility into trajectory optimization and legged locomotion planning. Retiming a nominal trajectory under predicted torque saturation is itself a mature sub-field, with roots the source planning document places in disturbance-observer-based path tracking under torque saturation in the early 2000s; we treat retiming (our Level 1, §V) as an established baseline mechanism, not a contribution, and flag this specific historical claim as not yet backed by a specific citation (see *Draft Status and Open Items*). A related task-space projection of torque limits for trajectory optimization on non-flat terrain is referenced in the source planning document without a specific citation and is flagged the same way.

For legged systems specifically, two prior works are the closest neighbors and have now both been read in full (see *Draft Status and Open Items*, item 4). Orsolino, Focchi, and colleagues' *Feasible Region* [5] computes a 2D convex set of CoM (x, y) positions that are simultaneously friction- and actuation-consistent, via an Iterative Projection algorithm intersecting a friction (support) region with a new actuation region derived from per-limb joint-torque wrench polytopes, and uses it online (100–133 Hz) to select footholds and CoM targets that maximize the region's area, validated on the HyQ quadruped carrying a 10 kg payload over a 0.22 m pallet. This is, to our knowledge, the closest prior work to projecting joint-torque limits through contact configuration into an actuation-aware planning quantity, and the source of the environment-conditioning idea in §IV for legged platforms. Two differences are load-bearing rather than incidental. First, the feasible region is explicitly a **quasi-static, geometric** set-membership test, not a predictive signal over a horizon: the paper states plainly that the construction "is only valid when the velocity of the robot's base is zero," and lists a *dynamic* extension (via dynamic wrench polytopes and CoM-acceleration terms) as future work, not something the paper does. The present work's certificate is instead a scalar margin evaluated over a receding horizon of *predicted* future states, contact, payload, and disturbance — the temporal, predictive dimension is precisely what is absent from the feasible region as published. Second, the feasible region's use in planning is a single mechanism (select the CoM/foothold target that lies inside, or nearest to, the region) with no analog of a graduated response or a condition for when that single mechanism has been exhausted; this paper's Level 0–4 hierarchy and Theorem 3's exhaustion condition have no counterpart there.

Acosta and Posa's perceptive mixed-integer footstep control (MPFC) for underactuated bipedal walking on rough terrain (*IEEE Transactions on Robotics*, 2025) [6] is the closest legged/terrain-aware actuation work published in this paper's own target venue, now read and differentiated in full rather than left as an open item. MPFC is a mixed-integer QP, solved online at over 100 Hz (mean 2.2 ms, 99.9th-percentile 7.7 ms, on hardware) on a reduced-order Angular-momentum Linear Inverted Pendulum (ALIP) model, that jointly optimizes discrete foothold selection among perceived convex terrain polygons, continuous footstep position and timing, and a single ankle-torque input, paired with a real-time terrain-segmentation perception stack (Stable Steppability Segmentation) engineered specifically against the temporal "flickering" instability of explicit plane segmentation. Three differences, again load-bearing. First, MPFC's discrete-and-continuous decision variables are jointly re-solved *every cycle*; there is no cheaper local-adaptation-first mechanism tried before a more expensive discrete (foothold/route) change is considered, and no analog of a checkable condition for when local adaptation has been exhausted — the present work's Level 0–4 escalation and Theorem 3 have no counterpart in MPFC's architecture, which is a different and equally valid design point (pay the joint-optimization cost every cycle, rather than pay a cheaper certificate check most cycles and a route-level cost only when needed). Second, MPFC's torque handling is narrow and downstream: ankle torque is one decision variable inside the reduced-order footstep optimization, and broader whole-body joint-torque realizability is delegated entirely to a separate 1 kHz operational-space-control QP with hard bounds — nothing in MPFC plays the role of a predictive, uncertainty-bounded margin computed jointly across actuators and fed back into the higher-level planner's decision, which is this paper's certificate's specific job. Third, and most fundamentally, MPFC's contribution is concentrated on $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}}$ under discrete, perceived terrain — *where it is geometrically and kinematically safe to step* — solved extremely well and validated extensively on hardware; this paper's contribution is specifically $\mathcal{F}_{\text{dyn}}$ — *whether the actuators can physically realize the motion once a location is chosen* — and MPFC does not represent that as a planning-time signal distinct from the ankle-torque decision variable itself. MPFC is also a systems/hardware paper with no formal theorem structure, which sharpens rather than undermines this paper's own (admittedly modest, §VI) theoretical contribution by comparison: there is no competing formal exhaustion or escalation condition to reconcile with.

E. Position of this work. No single piece of prior art we have located combines (i) a unified failure taxonomy spanning geometric and physical failure modes, (ii) a hierarchical adapt-then-reroute-then-fallback response with an explicit necessity condition for route-level change, (iii) a certificate jointly conditioned on predicted contact, payload, terrain, and disturbance, and (iv) demonstrated generality across a fixed-base manipulator and, pending resourcing, legged platforms. Each individual mechanism, however, is independently well established, and our contribution claims are scoped accordingly (§I-A, and see *What This Paper Does Not Claim* below).

III. Problem Formulation

Let $x_k = (q_k, \dot{q}_k)$ be the robot state at discrete time k, and let $E_{k:k+N}$ denote predicted environment information over a horizon of length N — terrain height and slope, expected contact locations and timing, friction, payload, and anticipated external disturbance or interaction force. A nominal motion planner $\mathcal{P}$ produces a candidate trajectory $u^{\text{nom}}_{0:N-1}$ (and the induced state sequence $x_{0:N}$) that is certified against two feasibility sets already representable by existing pipelines:

$$\mathcal{F}_{\text{kin}}(x) \quad \text{(kinematic feasibility: velocity, acceleration, joint-limit bounds)},$$
$$\mathcal{F}_{\text{obs}}(x, E) \quad \text{(collision/obstacle feasibility, given predicted environment } E\text{)}.$$

We introduce a third set, not represented by conventional pipelines: $\mathcal{F}_{\text{dyn}}(x, E)$, the set of states x at which the actuator command required to realize the requested motion under E lies within actuator limits.

The joint safe set the planner should certify is

$$\mathcal{F}_{\text{safe}} = \mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}} \cap \mathcal{F}_{\text{dyn}}.$$

Existing planning pipelines certify $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}}$ at planning time and discover violations of $\mathcal{F}_{\text{dyn}}$ only at execution time, typically through actuator saturation, tracking-error growth, or, on a floating-base or legged platform, loss of balance. The central problem this paper addresses is: *under what conditions can $\mathcal{F}_{\text{dyn}}$-membership be predicted ahead of execution, and how should the planner respond when prediction indicates future loss of membership?*

This paper's contribution and its evidence are both scoped to $\mathcal{F}_{\text{dyn}}$ specifically, not to certifying $\mathcal{F}_{\text{safe}}$ as a whole: the certificate, hierarchy, and theoretical results throughout are about physical (actuator) realizability, and the reduced-order verification in §IX runs with collision checking out of scope entirely (§IX-F).

$\mathcal{F}_{\text{obs}}$ is included in the joint safe-set definition above because a complete planner would need to certify it, and because MoveIt 2's OMPL layer already does — this paper does not repeat or extend that part, and reports $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{dyn}}$-level evidence, not $\mathcal{F}_{\text{safe}}$-level evidence. We use *physical realizability* for the actuator/torque/payload/contact concern this paper is actually about, and reserve *(overall) motion safety* for the full $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}} \cap \mathcal{F}_{\text{dyn}}$ claim this paper does not make.

We assume bounded model uncertainty (bounded error between the dynamics model used for prediction and the true plant) and bounded external/contact-force prediction error over the horizon; both assumptions are stated explicitly wherever they are used and are not claimed to hold unconditionally (see the assumption-scoped statement of Theorem 1, §VI).

We also assume, throughout, that actuator margin is generally an asset to be preserved rather than a resource the task is meant to spend: $\mathcal{F}_{\text{dyn}}$-membership is defined with a strictly positive safety threshold m_{safe}, so a trajectory that intentionally operates at or near the actuator limit is treated identically to one that is about to lose control authority unintentionally — both register as $m_{\text{phys}} < m_{\text{safe}}$. This is a real scope boundary, not a detail: minimum-time trajectories under torque constraints are bang-bang almost everywhere by construction (classical time-optimal control results, e.g. Bobrow et al.; Pontryagin's maximum principle), so a genuinely time-optimal reference has near-zero or negative margin by design, and this architecture would retime it away from the optimum it was constructed to reach. §XI states this as an explicit non-claim.

IV. Predictive Physical-Realizability Certificate

For the nominal predicted trajectory $\mathbf{x}_{0:N}, \mathbf{u}_{0:N}$, define the per-joint, per-horizon-step torque margin

$$m_{\tau,i}(k+j) = \tau_{i,\max} - |\tau_i(k+j)|,$$

and its uncertainty-tightened form, using a predicted torque estimate $\hat{\tau}_i$ and an explicit uncertainty bound $\Delta\tau_i$ that accounts for model mismatch and contact/disturbance prediction error,

$$m_{\tau,i}^{\text{robust}}(k+j) = \tau_{i,\max} - |\hat{\tau}_i(k+j)| - \Delta\tau_i(k+j).$$

The aggregate horizon certificate — the single scalar we carry into the theoretical results of §VI — is

$$m_{\text{phys}}(k) = \min_{j=0,\ldots,N} \; \min_i \; m_{\tau,i}^{\text{robust}}(k+j).$$

Three complementary signals are reported as experimental diagnostics rather than carried into the formal results, to keep the required proof burden proportional to a 3–4-contribution paper: directional authority $\alpha_{\text{dir}}(k+j)$ (remaining control/acceleration authority in the direction the current task demands, useful for distinguishing "some margin remains, but not in the direction needed" from an aggregate scalar); time-to-loss $T_{\text{loss}} = \min\{\, j\Delta t : m_{\text{phys}}(k+j) \le 0\,\}$; and cumulative risk $J_{\text{phys}} = \sum_j \phi(m_{\text{phys}}(k+j))$ for a chosen penalty function ϕ.

Environment conditioning. The certificate is a function of predicted environment and contact information as well as state:

$$(E_{k:k+N},\, x_k,\, u^{\text{nom}}_{k:k+N}) \;\longrightarrow\; \tau_{k:k+N} \;\longrightarrow\; m_{\text{phys}}.$$

This is the step that distinguishes the contribution from a torque-limit checker: a trajectory can be entirely collision-free and still fail $\mathcal{F}_{\text{dyn}}$ because the *predicted* terrain, contact, or payload state along the route demands more torque than is available, i.e.

$$\text{collision-free} \;\not\Rightarrow\; \text{physically realizable.}$$

A representative (illustrative, not yet run — see *Draft Status and Open Items*) instance: on flat ground a legged platform's ankle-torque margin is $m_{\text{ankle}} = 0.42$; the same nominal footstep trajectory into a predicted shallow hole yields $m_{\text{ankle}} = 0.05$; continuing the unmodified nominal trajectory would drive $m_{\text{ankle}} < 0$ at the moment of contact. The certificate exposes this before the step is taken, giving the planner the opportunity to adjust footstep placement, timing, or center-of-mass trajectory rather than discovering the deficit at contact.

V. Feedback Planning Architecture

A. Failure taxonomy

We retain the geometric/topological failure signal from prior work in this line ($\sigma_1 \vee \sigma_2 \vee \sigma_3$: QP infeasibility, planner stagnation, and prediction/Jacobian linearization drift, respectively) and add a physical-realizability signal:

$$\sigma_{\text{geo}} = \sigma_1 \vee \sigma_2 \vee \sigma_3, \qquad \sigma_4 = \left[\, m_{\text{phys}}(k+j) < m_{\text{safe}} \text{ for some } j \in [0, N] \,\right].$$

The two signals demand structurally different responses, and σ_4 must not be collapsed into the existing geometric-escape mechanism as though it were a fourth instance of the same trigger:

$\sigma_{\text{geo}} \Rightarrow$ **change WHERE**,

$\sigma_4 \Rightarrow$ **change HOW first, change WHERE only if adaptation cannot restore feasibility.**

The reason this distinction matters architecturally: a geometric escape mechanism retargets the route (h_{esc} in the underlying QP) while leaving the kinematic feasible set unchanged. This is the correct response to a topological trap, but it does nothing for σ_4: if m_{phys} has collapsed because the *current* route demands more torque than is available, retargeting within the same route's local neighborhood does not necessarily restore torque margin, and only retiming, reshaping, or a genuine route-level change can.

B. Hierarchical response

- **Level 0 — Normal execution.** If $m_{\text{phys}}(k) \geq m_{\text{safe}}$, execute the nominal trajectory unmodified.
- **Level 1 — Retiming.** The route remains dynamically realizable, but the nominal timing over-demands the actuators: reparameterize $q(s) \to q(t_{\text{new}})$ with reduced velocity/acceleration along the same geometric path. This mechanism is a mature baseline (§II-D), not a contribution of this paper.
- **Level 2 — Trajectory reshaping.** Modify the acceleration profile, center-of-mass motion, contact timing, or local target while remaining on the same route, to restore $m_{\text{phys}} > 0$ without changing the route. The reshape problem is posed as a QP over per-step state variables $(q_j, \dot{q}_j, \ddot{q}_j)$, anchored at the true current state via double-integrator integration constraints (so the certificate evaluated over its output certifies a trajectory the system can actually be commanded to follow, not just an acceleration sequence in isolation), with objective

$$w_a \sum_j \|\ddot{q}_j - \ddot{q}_j^{\text{nom}}\|^2 + w_q \sum_j \|q_j - q_j^{\text{nom}}\|^2 + w_{\dot{q}} \sum_j \|\dot{q}_j - \dot{q}_j^{\text{nom}}\|^2,$$

  i.e. the closest dynamically-feasible trajectory to the one the planner originally intended, not the smallest-acceleration one — deliberately, since minimizing raw $\sum \|\ddot{q}\|^2$ has no reason to prefer a result close to the planner's original intent.

- **Level 3 — Rerouting/replanning.** If no route-level regeneration along the current route restores realizability — neither retiming nor reshaping, tried in that order (§IX-H) — the route itself must change: $p \to p'$. Theorem 3 (§VI) formally covers only the retiming-adaptation margin $m_1^\star(p) < m_{\text{safe}}$ (a sufficient, checkable condition for the idealized $\mathcal{T}_{\text{dyn}}(p) = \emptyset$); folding the implemented, empirically-verified reshape check into a formal, extended adaptation class $m^\star(p)$ is not yet done and is noted as such where it matters (§V-C, §IX-H).
- **Level 4 — Safe braking/fallback.** If no feasible continuation exists within the available reaction horizon, invoke a braking or terminal-safe-set policy. This level exists so the architecture does not implicitly assume every failure is solvable by further replanning.

C. The Level 3 mechanism (open item)

Theorem 3 (§VI) answers *when* the architecture should stop attempting local adaptation and invoke Level 3, via the checkable condition $m_1^\star(p) < m_{\text{safe}}$. It does not answer *what the rerouter should then do*, and that remains a genuinely open design question rather than something to assume away. Simply invoking "global replanning or the existing geometric-escape mechanism" is not yet justified: the existing geometric-escape mechanism selects an escape target to restore topological/geometric clearance; it has no explicit notion of torque margin, and there is no a priori reason a geometrically-motivated escape target also satisfies lower dynamic demand. Two directions close this gap, neither implemented as of this draft: (a) show that geometric-escape-selected targets are more dynamically realizable than the original route under stated conditions, or (b) give σ_4-triggered rerouting an explicit objective distinct from the geometric-escape mechanism's — for example, selecting among candidate routes P by $\min_P J_{\text{geometric}}(P)$ subject to $m_{\text{phys}}(P) \geq m_{\text{safe}}$ (a physical-feasibility *constraint* on route selection, not merely a preference), or constraining the geometric-escape target search to $\mathcal{F}_{\text{dyn}}$-feasible candidates directly. Either would also let Theorem 3's adaptation class be extended from $\mathcal{A}_1(p)$ (retiming only) to include reshaping and, eventually, route selection itself, closing more of the gap to the idealized $\mathcal{T}_{\text{dyn}}(p) = \emptyset$ condition discussed in Theorem 3's proof. This step is currently assumed, not shown, and must be closed before the Level 0–4 hierarchy's Level 3 mechanism (as opposed to its triggering condition) can be considered scientifically justified rather than a placeholder.

Update (§IX-H): the CODE-level version of this extension — trying reshape as route-level regeneration, alongside retiming, before ever escalating to reroute — is now implemented and empirically verified: reshape genuinely cannot resolve either of this benchmark's existing reroute-required scenarios (confirmed against an independent solver), and a scenario exists where it succeeds where retiming cannot, by reallocating timing within a fixed route duration rather than by geometric avoidance. This is *not* the same as extending Theorem 3's formal statement to a richer $m^\star(p) = \sup(\mathcal{A}_1(p) \cup \mathcal{A}_2(p))$ — no monotonicity-style hypothesis analogous to Theorem 3's has been stated or checked for the reshape class, so the code now does more than the theorem formally covers, an asymmetry disclosed here rather than papered over. The harder question this section raises — what the rerouter should do, i.e. (a)/(b) above — remains fully open; folding reshape into the pre-execution class does not touch it, since reshape never picks a new geometric target at all.

VI. Theoretical Analysis

Throughout, we use ξ_k generically for a tracking-error or deviation state where relevant to a specific result, and reuse the sets $\mathcal{F}_{\text{kin}}$, $\mathcal{F}_{\text{obs}}$, $\mathcal{F}_{\text{dyn}}$, $\mathcal{F}_{\text{safe}}$ from §III.

Theorem 1 (Realizability certificate). *Under bounded model uncertainty and bounded external/contact-force prediction error over the horizon, if $m_{\text{phys}}(k+j) \geq 0$ for all $j \in [0, N]$, then the predicted nominal trajectory satisfies the actuator constraints over the horizon.*

Proof strategy. This is definitional once the assumptions are made explicit: $m_{\text{phys}}(k+j) \geq 0$ means, by construction of $m_{\tau,i}^{\text{robust}}$, that for every joint i and every horizon step j, $\tau_{i,\max} - |\hat{\tau}_i(k+j)| - \Delta\tau_i(k+j) \geq 0$. If the true torque $\tau_i(k+j)$ satisfies $|\tau_i(k+j) - \hat{\tau}_i(k+j)| \leq \Delta\tau_i(k+j)$ — the bounded-uncertainty assumption, restated as an explicit per-step guarantee rather than an unconditional claim — then $|\tau_i(k+j)| \leq |\hat{\tau}_i(k+j)| + \Delta\tau_i(k+j) \leq \tau_{i,\max}$ follows directly, for every joint and every horizon step, which is the statement that the actuator constraints hold over the horizon. The theorem is exactly as strong as the uncertainty bound $\Delta\tau_i$ used to construct it, and it certifies the constraint-inactive predicted trajectory *as predicted* — it does not, by itself, say anything about closed-loop behavior once feedback, replanning, or an inaccurate $\Delta\tau_i$ are in play; that is addressed separately by the assumptions attached to Theorems 2 and 3 and is not claimed here.

Theorem 2 (Realizability-preserving adaptation). *Suppose the nominal route admits at least one trajectory with $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}} \cap \mathcal{F}_{\text{dyn}} \neq \emptyset$ (an explicit precondition). If the local adaptation problem (Level 1/2, §V-B) includes the predictive dynamic-feasibility constraint and is feasible, the resulting trajectory remains geometrically feasible while restoring actuator realizability, without necessarily changing the global route.*

Proof strategy. By the stated precondition, a nonempty target set exists within the current route's neighborhood. The Level 1/2 adaptation problem is posed as a constrained optimization over retiming/reshaping parameters subject to $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}} \cap \mathcal{F}_{\text{dyn}}$ jointly, rather than $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}}$ alone as in a conventional retiming stage; if this constrained problem returns a feasible solution, that solution is a certificate of membership in the joint set by construction of the problem, and membership in $\mathcal{F}_{\text{kin}} \cap \mathcal{F}_{\text{obs}}$ in particular means the result is unchanged in its geometric route. The result is essentially an existence statement conditioned on solver feasibility, not a claim that the adaptation problem is always feasible when the precondition holds — the precondition guarantees *some* feasible trajectory exists along the route, not that the specific retiming/reshaping parameterization used by Level 1/2 is expressive enough to reach it. Closing that expressiveness gap for a specific parameterization is implementation-dependent and is addressed, for the retiming parameterization specifically, by Theorem 3 below.

Theorem 3 (Sufficient, checkable condition for rerouting necessity). *Let p be a route and ξ_0 the nominal trajectory the local planner produces along it. For a retiming factor λ in the allowed range $\Lambda = [1, \lambda_{\max}]$, let ξ_λ denote ξ_0 time-reparameterized by λ (Level 1, §V-B), and define the* retiming-adaptation class $\mathcal{A}_1(p) = \{\xi_\lambda : \lambda \in \Lambda\}$ *— the set of trajectories the architecture's pre-execution route decision actually searches before invoking Level 3 — and the* retiming-adaptation margin $m_1^\star(p) = \sup_{\lambda\in\Lambda} m_{\text{phys}}(\xi_\lambda)$*, the aggregate horizon certificate (§IV) evaluated over the whole route under ξ_λ. Under the assumption that $\lambda \mapsto m_{\text{phys}}(\xi_\lambda)$ is non-decreasing on Λ (retiming monotonicity — stated as an explicit hypothesis, in the manner of Theorem 1's uncertainty bound, not derived from first principles for the general nonlinear case; see proof), if $m_1^\star(p) < m_{\text{safe}}$, then no trajectory in $\mathcal{A}_1(p)$ restores the certificate, and retiming is genuinely exhausted, not merely assumed exhausted: escalation past Level 1 to the architecture's next adaptation mechanism is warranted. This theorem covers only the retiming class $\mathcal{A}_1(p)$; the implemented pre-execution decision (§V-C, §IX-H) tries Level 2 reshaping next and only escalates to Level 3 rerouting if reshaping also fails to restore the certificate — a claim this theorem does not itself formalize (see below and §V-C).*

This is a sufficient, not necessary, condition for the idealized statement that $\mathcal{T}_{\text{dyn}}(p) = \emptyset$ (the set of all *dynamically realizable trajectories along p, of which the observation "$\mathcal{T}_{\text{dyn}}(p) = \emptyset \Rightarrow$ no modification staying on p is dynamically realizable" is immediate from the definition, and is not by itself a substantial claim — an earlier version of this theorem stated exactly that idealized observation and presented it as the main result, which is the "almost tautological" framing a T-RO reviewer would reasonably object to). Since $\mathcal{A}_1(p) \subseteq \mathcal{T}_{\text{dyn}}(p)$ whenever any element of $\mathcal{A}_1(p)$ happens to be dynamically feasible, $m_1^\star(p) < m_{\text{safe}}$ does not imply $\mathcal{T}_{\text{dyn}}(p) = \emptyset$: a dynamically realizable trajectory along p may exist outside $\mathcal{A}_1(p)$, reachable only through a richer adaptation mechanism the current pre-execution decision does not attempt — most directly, Level 2 reshaping, which is available online (§V-B) but not yet folded into this route-level check. Closing that gap by defining a joint retiming-and-reshaping class $\mathcal{A}(p) \supseteq \mathcal{A}_1(p)$ is identified as a concrete next step, not yet implemented (§V-C,* Draft Status and Open Items*).*

Proof. Under the monotonicity assumption, $\sup_{\lambda\in\Lambda} m_{\text{phys}}(\xi_\lambda) = m_{\text{phys}}(\xi_{\lambda_{\max}})$, so $m_1^\star(p) < m_{\text{safe}}$ is equivalent to $m_{\text{phys}}(\xi_{\lambda_{\max}}) < m_{\text{safe}}$, checkable with a single evaluation at the boundary of Λ (in the implementation, a bisection search over Λ that only needs to rule out $\lambda_{\max}$ to certify infeasibility of the whole class). Monotonicity itself is not proven for the fully general nonlinear case: under uniform time-dilation by λ, velocity and acceleration along the path scale as $1/\lambda$ and $1/\lambda^2$, so the torque contributions from the mass-matrix and Coriolis/centrifugal terms shrink in magnitude as λ grows, while the gravity and any velocity-independent external-force contribution are unchanged by λ — under this structure, slowing down cannot increase the magnitude of a torque contribution that shares gravity's sign, but the Lemma below states precisely when the assumption can fail, rather than leaving it as an unquantified hedge. Given the assumption, $m_{\text{phys}}(\xi_\lambda) \le m_{\text{phys}}(\xi_{\lambda_{\max}}) < m_{\text{safe}}$ for every $\lambda \in \Lambda$, i.e. no element of $\mathcal{A}_1(p)$ restores the certificate, which is exactly the condition under which the implemented pre-execution decision correctly escalates past retiming, to reshaping and then rerouting if reshaping too proves insufficient. ■

Lemma (per-joint sign condition for retiming monotonicity). *Fix a route p and a path fraction s, and let $\tau_i(\lambda)$ be the joint-i torque required at that fraction under ξ_λ. By the manipulator equation, for a possibly position-dependent (not wall-clock-dependent — see below) external force $F_{\text{ext}}(q)$,*

$$\tau_i(\lambda) = \frac{A_i}{\lambda^2} + B_i, \qquad A_i \equiv M_i(q)\,\ddot{q}_0(s) + C_i(q, \dot{q}_0(s))\,\dot{q}_0(s), \qquad B_i \equiv g_i(q) + J_i(q)^\top F_{\text{ext}}(q),$$

where A_i (the inertial and Coriolis/centrifugal contribution, evaluated once at $\lambda = 1$'s velocity/acceleration at this path fraction) scales as $1/\lambda^2$ under uniform time-dilation — since $\dot{q}_\lambda = \dot{q}_0/\lambda$, $\ddot{q}_\lambda = \ddot{q}_0/\lambda^2$, and $C(q,\dot{q})\dot{q}$ is bilinear in $\dot{q}$ — while B_i (gravity plus the velocity-independent external-force contribution) is unchanged by λ. If, for every joint i and every horizon/route step evaluated, either $A_i = 0$ or $\operatorname{sign}(A_i) = \operatorname{sign}(B_i)$, then $\lambda \mapsto m_{\text{phys}}(\xi_\lambda)$ is monotonically non-decreasing on Λ, and Theorem 3's boundary-only check correctly computes $m_1^\star(p) = m_{\text{phys}}(\xi_{\lambda_{\max}})$.

Proof. When $\operatorname{sign}(A_i) = \operatorname{sign}(B_i)$ (or $A_i = 0$), $|\tau_i(\lambda)| = |A_i|/\lambda^2 + |B_i|$ is strictly decreasing in λ for $A_i \neq 0$ and constant for $A_i = 0$, so the per-joint, per-step margin $m_{\tau,i}(\lambda) = \tau_{i,\max} - |\tau_i(\lambda)| - \Delta\tau_i$ is non-decreasing on Λ. The aggregate $m_{\text{phys}}(\xi_\lambda) = \min_{i,\text{step}} m_{\tau,i}(\lambda)$ is a minimum of a finite collection of such functions, hence itself non-decreasing on Λ. ■

When the condition fails, and why it is not a corner case. If instead $\operatorname{sign}(A_i) \neq \operatorname{sign}(B_i)$ at some binding joint — the inertial/Coriolis torque *partially unloads* the gravity/external-force torque, e.g. decelerating through the second half of a downward swing, or a Coriolis coupling from another joint's motion that happens to counteract gravity at that configuration — then $\tau_i(\lambda)$ crosses zero at $\lambda_i^\star = \sqrt{-A_i/B_i}$ (when real and in Λ), and $m_{\tau,i}(\lambda)$ has an *interior* maximum at $\lambda_i^\star$ rather than at $\lambda_{\max}$: the boundary-only check can then report $m_1^\star(p) < m_{\text{safe}}$ (retiming "exhausted") when an interior λ would in fact have cleared m_{safe}. A random search over 3000 scenarios on this platform's implementation (§IX) found this failure mode in 23 cases (~0.8%) — e.g. one instance with $m_{\text{phys}}(\lambda_{\max}) = 1.66 < m_{\text{safe}} = 2.0$ while $m_{\text{phys}}(1.55) = 2.86$ — confirming the condition fails often enough to matter, not merely in principle. The Lemma's condition is checkable at negligible marginal cost, since A_i and B_i are already recoverable from one extra zero-velocity/zero-acceleration inverse-dynamics evaluation at the same configuration used for $\tau_i(1)$ (B_i directly, $A_i = \tau_i(1) - B_i$), so the implementation uses it as a certificate for the fast boundary-only search before trusting it, rather than trusting it unconditionally: when the sign condition holds, the boundary check alone is used (unchanged cost); when it does not, a dense scan of Λ with local refinement around the first crossing is used instead (§IX-H, roughly 3–80ms in the cases tested, against the 1–3ms boundary-only cost) — a resolution-limited but checkable and tunable fallback, not a silent assumption. An exact, closed-form alternative to the dense scan — evaluating the finite candidate set $\{1, \lambda_{\max}\} \cup \{\lambda_i^\star : \lambda_i^\star \in \Lambda\}$ directly from the decomposition above, rather than sampling a grid — is identified as the natural next step and is being developed for the narrower saturation-certificate paper this Lemma is intended to feed (`monotonicity_lemma_draft.md` in the repository; *Draft Status and Open Items*, item 6), but is not yet implemented or claimed here.

This is empirically consistent with the one negative case already tested for it (§IX): a pure static (gravity-only) torque deficit, where the velocity/acceleration-driven contribution is zero throughout ($A_i \equiv 0$ for every joint, the case the Lemma's condition holds trivially) and retiming provably cannot help regardless of λ, correctly causes $m_1^\star(p) < m_{\text{safe}}$ and a fall-through past Level 1 — not a proof that monotonicity holds in general, but a case where the Lemma's own justification is exact rather than approximate. Both results actually cited in this paper (§IX-B's flagship scenario and §IX-G's Experiment 7) were checked directly against the Lemma's sign condition rather than assumed: the flagship scenario is genuinely monotonic throughout (verified by dense scan), and Experiment 7 is technically non-monotonic but the interior deviation (~ 0.5) is negligible against how far below m_{safe} it stays throughout (~ -20) — neither reported conclusion changes. A related, separate subtlety worth stating explicitly: retiming ξ_0 to ξ_λ changes which predicted environment sample $E(k+j)$ a given configuration is paired with whenever E is genuinely a function of wall-clock time, $E = E(t)$

(an independently time-driven disturbance, unaffected by how fast the robot moves through it) rather than a function of configuration or path progress, $E = E(q)$ (e.g. a terrain/contact feature encountered at a specific point along the route, however long that takes to reach) — the case the Lemma above is scoped to. For $E(q)$-type environments, retiming leaves the configuration-to-environment pairing unchanged and $m_1^\star(p)$ is evaluated consistently; for $E(t)$-type environments, slowing down changes which $E(k+j)$ a given point on the route encounters, so $m_1^\star(p)$ implicitly assumes the SAME predicted $E(t)$ schedule applies regardless of λ — correct only if the environment prediction is itself re-sampled at the retimed schedule, which this architecture does (the implementation's force/environment interface takes (t, q) jointly and is re-sampled at every λ tested, so this is handled correctly in the current implementation) but is not yet stated as a distinct case in the theorem itself.

Theorem 4 (Recursive feasibility / safe fallback) — deferred. A terminal safe set $\mathcal{X}_f$ such that every $x_k \in \mathcal{X}_f$ admits a bounded safe braking policy satisfying actuator and collision constraints, with the recursive-feasibility property $x_k \in \mathcal{X}_f \Rightarrow x_{k+1} \in \mathcal{X}_f$, would strengthen Level 4 (§V-B) from a heuristic fallback into a formally guaranteed one. We have not constructed such a set or proved this property for the present architecture, and we do not claim it here. Per the source planning document's own guidance, this result is attempted only if the required assumptions can be established without becoming so restrictive as to be vacuous for the platforms in §VIII; if that turns out not to be achievable, Theorem 4 should be omitted from any submission-ready version of this paper and Level 4 presented as an engineered, empirically-validated fallback rather than a formally certified one. This is stated as an open item, not resolved.

Computational structure. A property we carry over unchanged from the underlying fixed-structure local planner, scoped explicitly to the certificate evaluation and Level 0–1: the prediction matrices and QP Hessian $(H, \Phi, \Gamma, C_{\text{kin}})$ remain fixed online; state- and environment-dependent information enters only through vector terms ($h(x, E)$, $d(x, E)$, margin vectors), the same mechanism already used for obstacle rows. Adaptive numerical values are not the same as an adaptive solver structure, and preserving this distinction is what keeps that part of the architecture real-time-capable as more predictive signals are added (§VII). This property is explicitly NOT claimed for Level 2 (reshaping): its QP has no analogous fixed, reusable structure in the current implementation, is rebuilt from scratch every call, and §IX-E/H measures the resulting real-time cost directly rather than assuming it away — a parametrized/warm-started reformulation is the natural fix but is not yet built (§IX-E). The real-time claim of this paper is therefore properly scoped to certificate evaluation and Level 0–1, not the full Level 0–4 hierarchy.

VII. MoveIt 2 Integration Architecture

We target MoveIt 2 Hybrid Planning as the integration interface for this architecture. A real integration now exists and is reported in §IX-I: C++ `LocalConstraintSolverInterface/TrajectoryOperatorInterface` plugins implementing B2/B3 against a real Franka FR3 kinematic/dynamic model, executed under MuJoCo. That platform currently exercises Levels 0/1/2/4 of the hierarchy below (retime, reshape, brake) plus the certificate itself; Level 3 (reroute, the geometric-escape mechanism $\sigma_1..\sigma_3$) has not yet been exercised on it — only on the reduced-order platform of §IX-B/G. What follows is stated as the target architecture this integration was built toward; §IX-I reports what has actually been run and measured, including what remains open. The integration preserves MoveIt 2's global planner (OMPL) unmodified and replaces only the local-planner component of MoveIt's Hybrid Planning interface, which is explicitly designed to accept custom global/local planner plugins and event-driven planning logic:

```
MoveIt Global Planner (OMPL)
        |
        v
  Global trajectory
        |
        v
Predictive-Realizability-Aware Local Planner
        |
```

```
    +-- fixed-structure local planner (nominal trajectory)
    +-- geometric-escape mechanism (sigma_1..sigma_3)
    +-- predictive physical-realizability certificate (sigma_4, Sec. IV)
    |
    v
Controller --> Robot --> state/contact/force feedback
```

The local-planner slot runs the fixed-structure planner at its native rate, evaluates m_{phys} over the receding horizon each cycle using the current state estimate and the latest available environment/contact prediction, and triggers the Level 0–4 response of §V based on σ_{geo} and σ_4. Because the certificate and the geometric-escape mechanism both enter the underlying QP only through vector terms rather than restructuring it, the additional real-time cost of evaluating m_{phys} each cycle is a torque-prediction pass over the horizon (an inverse-dynamics evaluation per step) plus a margin computation, not a change in the QP's sparsity structure or size. §VIII-J reports the measured cost of this addition rather than asserting it is negligible.

VIII. Experimental Evaluation — Proposed Benchmark Design

This section remains, in the main, a design specification rather than a results section: every number below is a planned parameter, not a measured outcome, for Experiments 4-7 and for the real-time/ablation protocol as a whole. That is no longer true of Experiments 1-3, however: §IX-I now reports real measured results for those three, on an actual MoveIt 2 Hybrid Planning / FR3-kinematics platform (simulated dynamics via MuJoCo, not the reduced-order proxy below), not merely a plan. §IX also reports a preliminary reduced-order check of the full design (certificate, hierarchy, ablations, real-time protocol, including Experiments 5-7's flagship-reroute analogs) on a smaller, faster-to-iterate 3-DOF platform; between the two, no experiment in this section has been run at both FR3 scale and in its full form specified here, and each should be read strictly as what it is (§IX-I: partial coverage at real scale; §IX-B–H: full coverage at reduced scale) rather than either one standing in for the complete study. All experiments in this section are scoped to what is reachable today (fixed-base manipulator, simulation with optional real-hardware validation); legged/humanoid extensions are described separately in §VIII-K and explicitly deferred pending the resourcing decision noted in *Draft Status and Open Items*.

A. Platform and baselines

Platform. Franka FR3 (or Panda), MoveIt 2, ROS 2, MuJoCo (or Isaac Sim) as the primary simulator, with optional real-hardware validation on the FR3 if available. This platform is chosen because it has mature MoveIt integration, well-characterized per-joint torque limits, and manageable payload/interaction-force experiment design — matching Phase I of the source planning document's platform staging.

Baselines and the system under test. - **B1 — MoveIt standard.** OMPL (global) → TOTG (time parameterization) → execution. No predictive feedback of any kind. - **B2 — MoveIt Hybrid Planning, reactive local planner.** OMPL (global) + a conventional reactive local planner that responds to *current*-state saturation (e.g., clips or re-scales the command when the current torque exceeds a limit) but does not predict. - **B3 — Proposed architecture.** OMPL (global) + fixed-structure local planner + geometric-escape mechanism + predictive physical-realizability certificate (this work).

B2 vs. B3 is the primary, fair comparison: both share the same global planner and the same local-planner slot in MoveIt's architecture, differing only in whether the local planner is reactive or predictive. B1 is retained as the unmodified-default reference point.

B. Experiment 1 — Baseline manipulation trajectory

A representative reach-and-place trajectory (pick-and-place style, no payload change, static obstacles) is executed under B1/B2/B3. Metrics: planning time, execution time, peak per-joint torque, minimum torque margin achieved during execution, count of declared-acceleration-limit violations, tracking error (RMS and peak), and minimum obstacle clearance. This experiment establishes that the proposed architecture does not regress ordinary-case performance relative to B1/B2 before any of the harder scenarios below are introduced. §IX-I reports a real result for this experiment on the FR3-scale MoveIt 2 platform.

C. Experiment 2 — Payload variation (authority loss without geometry change)

The identical geometric trajectory from Experiment 1 is re-executed at a sequence of payload masses spanning the manipulator's rated capacity (e.g., 0%, 40%, 70%, 90%, 100%, 110% of rated payload — the 110% point deliberately probes just past the nominal rating). Because the geometric path and time law are held fixed, this experiment isolates the central claim that identical geometry does not imply identical physical feasibility. Primary metric: at which payload level does each baseline first exhibit a torque-limit violation or tracking-error growth, versus at which payload level B3's certificate first reports $m_{\text{phys}} < m_{\text{safe}}$ and triggers Level 1/2 adaptation — the gap between these two payload levels (if any) is the quantity of interest, not a single pass/fail outcome. §IX-I reports a real result for this experiment on the FR3-scale MoveIt 2 platform, with an important framing caveat: B2 and B3 use different trigger thresholds by construction, so the gap is best read as an activation-threshold separation, not a "prediction lead time" (that claim is reserved for Experiment 3, where the comparison is genuinely about lookahead).

D. Experiment 3 — External interaction force (detection lead time)

During execution of the Experiment 1 trajectory, an external wrench $F_{\text{int}}(t)$ is applied at the end-effector or a specified link (e.g., a scripted push profile with a ramp onset, magnitude swept across a range informed by the robot's rated payload/force capacity). Compared: B1 (no handling), B2 (reactive, current-state saturation handling only), B3 (proposed). Primary metric:

$$T_{\text{warning}} = T_{\text{failure}} - T_{\text{detection}},$$

the lead time between when the system first has actionable information that a failure is coming and when the failure would actually occur under an unmodified nominal trajectory. $T_{\text{warning}} = 0$ or undefined for B1 by construction (no detection mechanism); the comparison of interest is B2 versus B3, since B2 detects only once torque is already at the limit while B3 predicts the crossing before it occurs. §IX-I reports a real result for this experiment on the FR3-scale MoveIt 2 platform, including a direct predicted-vs-observed accuracy check for the certificate's own future-margin prediction, not only whether it fires ahead of B2.

E. Experiment 4 — Predicted environmental/contact transition (contact-stiffness/payload-step discontinuity)

The source planning document's original terrain experiment (flat/slope/hole/step) presumes a legged or mobile platform that is not part of the reachable Phase I scope (see §VIII-K and *Draft Status and Open Items*). We substitute a fixed-base-manipulator instance of the same general $E(q)/E(q,t)$ environment-conditioning framework (§IV) — this is not terrain, and is not presented as one, but is the same phenomenon the eventual legged terrain case would exercise: a trajectory can be collision-free while the *predicted* environment along the route changes what torque is required. The instance used here, without requiring a terrain rig, is a step change in end-effector contact stiffness or an abrupt, scripted payload change (e.g., a simulated hand-off or a contact transition from free space to a stiff surface) at a known point along an otherwise fixed geometric trajectory. The environment-prediction signal $E_{k:k+N}$ is the known upcoming contact/stiffness transition; the comparison is whether B3's certificate anticipates the transition and adapts (Level 1/2) before the transition occurs, versus B1/B2 discovering the transition's effect only once already in contact. This experiment demonstrates the environment-conditioning contribution (§IV) without deferring it entirely to Phase II; a future legged instance of the same framework would naturally read as terrain → contact force → m_{phys}. On the FR3-scale MoveIt 2 platform this experiment's implementation is complete but not currently runnable, blocked on a platform-level limitation unrelated to the certificate architecture; §IX-I reports what blocks it and what was learned diagnosing the blocker.

F. Experiment 5 — Flagship 1 (configuration/payload-induced): geometrically feasible, dynamically infeasible route

Construct two candidate routes between the same start and goal configuration: P_A, shorter in path length but passing through a persistently high-static-torque, gravity-disadvantaged configuration (e.g., an outstretched pose under payload, where gravity alone demands large joint torque irrespective of speed); and P_B, longer but remaining in a well-conditioned, low-static-demand "tucked" region throughout, under the same payload. A conventional planner selects P_A on path-length grounds alone. This experiment is FR3-realizable without any terrain rig — the "environment" here is the manipulator's own configuration-dependent conditioning under a fixed payload, which is already fully characterized by the FR3's kinematics and dynamics model. Expected comparison: B1/B2 select and attempt P_A, discovering the dynamic infeasibility during execution (torque saturation, tracking failure, or a stall); B3's certificate predicts $m_1^\star(P_A) < m_{\text{safe}}$ (retiming cannot restore margin) before committing to P_A and triggers Level 3 rerouting to P_B. This is the experiment most directly validating Theorem 3, and is proposed as one of the paper's two flagship results, alongside Experiment 7 (§VIII-L) — this one isolating a configuration/payload-driven deficit, Experiment 7 isolating an environment-driven one, so together they demonstrate that the same rerouting mechanism responds to genuinely different causes, not just one scripted scenario. This gravity-disadvantaged-versus-tucked framing (rather than an earlier draft's near-singular-wrist example) is deliberate: a route through a near-singular configuration risks reading as a kinematic-conditioning problem rather than a physical-realizability one specifically, whereas static torque demand at a well-conditioned pose makes the physical (not merely kinematic) nature of the deficit unambiguous. It is also not a hypothetical fix — it is exactly the construction already implemented and verified in §IX-B's reduced-order platform (the "outstretched" versus "tucked" poses reported there), so this section's design spec now matches what was actually built rather than lagging behind it.

G. Experiment 6 — Adaptation-versus-rerouting continuum

A single scenario family is parameterized by severity (e.g., a swept payload or interaction-force magnitude) to produce four regimes, directly validating the Level 0–4 hierarchy of §V-B rather than only its endpoints: - **Case A (small margin loss):** Level 1 retiming alone is sufficient to restore $m_{\text{phys}} \geq m_{\text{safe}}$. - **Case B (moderate loss):** retiming alone is insufficient; Level 2 reshaping is required and sufficient. - **Case C (severe loss):** no retiming or reshaping along the current route suffices; Level 3 rerouting succeeds. - **Case D (no safe route within the reaction horizon):** Level 4 braking/fallback is invoked. Metric: at each swept severity level, which hierarchy level actually resolves the scenario, compared against which level the certificate *predicts* will be sufficient — the interesting failure mode to look for is the certificate under- or over-predicting the required response level, not merely whether the scenario is eventually resolved.

H. Ablations

- **A1 — No predictive feedback.** Equivalent to B1.
- **A2 — Current-state saturation only, no prediction.** Equivalent to B2.
- **A3 — Prediction without planner feedback.** The certificate is computed and logged but not connected to the planner response (detects, does not adapt). Isolates whether prediction alone (as a monitoring signal, e.g. for an operator alert) has value distinct from acting on it.
- **A4 — Prediction + adaptation, no rerouting.** Levels 0–2 only; Level 3/4 disabled. Isolates the marginal value of rerouting specifically, expected to show degraded performance on Experiment 5/6-Case-C specifically.
- **A5 — Full proposed system.** Levels 0–4 (equivalent to B3).

I. Metrics (full set)

Task success rate; collision rate; minimum actuator margin achieved; count of saturation samples; tracking error (RMS, peak); minimum obstacle clearance; replanning count; replanning latency; local-planner per-cycle computation time; global-planner invocation rate; and **conservatism** — the fraction of executions in which the system triggers Level 2/3/4 response despite the nominal trajectory being, in fact, realizable (a false-positive rate on σ_4). Conservatism is reported alongside every safety-improvement claim in this paper, not as a secondary metric: an overly conservative certificate can trivially eliminate every failure mode above by triggering fallback constantly, which would make every other metric look good while the system is useless for its task. Any headline claim of reduced failure rate must be reported together with the corresponding conservatism figure. This metric is one-sided by construction: it bounds false positives (unnecessary triggers) but not false negatives (a real violation the certificate fails to catch before it occurs, e.g. from an inaccurate $\Delta\tau_i$ or an environment prediction error beyond what was assumed). A false-negative rate is not currently measured anywhere in this benchmark design and would need its own experiment; this is a gap, not an oversight to be inferred silently.

J. Real-time performance

Local-planner per-cycle computation time is measured directly (not assumed) for B2 and B3 under Experiment 1's nominal trajectory and under Experiment 5's gravity-disadvantaged stress case, reporting mean, p95, and maximum, against the local planner's target cycle rate. Whether the added certificate evaluation fits the real-time budget at the stress case specifically — not only the nominal case — is treated as an open empirical question the experiment must answer, not a design assumption.

K. Deferred: legged/humanoid extensions (Phase II/III)

Uneven-terrain, footstep-adaptation, and humanoid-balance experiments matching the original terrain scenario in the source planning document remain valuable future extensions of this benchmark suite but are explicitly **not** part of the Phase I reachable scope, pending the resourcing assessment in *Draft Status and Open Items*: no legged/humanoid simulation or hardware pipeline currently exists in this line of work, and building a floating-base, contact-scheduled dynamics model is architecturally distinct from the fixed-base manipulator work above.

L. Experiment 7 — Flagship 2 (environment-induced): environment-conditioned rerouting

Experiment 5 (§VIII-F) demonstrates rerouting driven by a configuration/payload deficit — the torque demand comes from the manipulator's own kinematics and inertia, not from anything in the environment. Experiment 4 (§VIII-E) demonstrates adaptation to a known environmental transition, but only ever gives the planner one route, so it cannot show rerouting *because of* that transition. This experiment closes that gap directly: two candidate routes P_A, P_B share the same start and goal configuration (as in Experiment 5); P_A's path enters a known region of environmental demand — reusing Experiment 4's scripted contact-stiffness model (a virtual surface exerting a penetration-proportional force) rather than introducing a new environment mechanism — while P_B's path avoids that region entirely. Distinguishing feature by design: P_A's deficit is to be verified as absent under the payload alone (a healthy static margin with the environmental force removed) and present only once the known force is included, isolating that this is the environment driving the reroute, not configuration or payload as in Experiment 5. As in Experiment 5, the force is a function of end-effector position (not of speed), so the deficit at the route's own zero-velocity/zero-acceleration via-point boundary is expected to be retiming-proof for the same reason Experiment 5's static deficit is, and Level 1 alone should not be expected to resolve it. Expected comparison: B1/B2 attempt P_A and discover the deficit once already in the field (torque saturation, tracking failure); B3, given the contact model at planning time as in Experiment 4 (`force_known_at_plan_time=True`), predicts P_A's margin before ever entering the field and reroutes to P_B.

IX. Preliminary Reduced-Order Verification

Since the benchmark design of §VIII was written, the mechanisms it specifies — the certificate, the Level 0–4 hierarchy, the ablation set, and the real-time protocol — have been implemented and run, not on the FR3-scale platform §VIII-A specifies, but on a smaller reduced-order system built to check whether the design behaves as intended before committing to that larger study. §IX-A through §IX-H report those results in full: certificate, hierarchy, ablations, real-time protocol, and both flagship reroute scenarios (Experiments 5 and 7's analogs). §IX-I then reports a second, independent line of evidence: real results from Experiments

1-3 run on the actual FR3-scale MoveIt 2 platform §VII describes, rather than the reduced-order proxy. The two are complementary, not overlapping — the reduced-order platform covers the full design breadth (including rerouting) at reduced physical scale, the FR3 platform covers real MoveIt 2/FR3 scale but only a subset of the design (Levels 0/1/2/4, not Level 3) — and neither alone establishes the complete §VIII specification. §VIII's own $\mathcal{F}_{\text{obs}}$/collision dimension and any legged/humanoid extension (§VIII-K) remain entirely unattempted in both.

A. Scope and platform

A 3-degree-of-freedom planar revolute arm, moving in a vertical plane so gravity contributes a real, configuration-dependent torque demand, with per-joint torque limits (87/87/12 Nm) chosen to be FR3-representative in scale though not in kinematic structure. Dynamics are computed by a physics engine (MuJoCo) rather than hand-derived, and were independently checked before any certificate result was trusted: the mass matrix is symmetric and positive-definite over random configurations, the static torque at a known configuration matches a closed-form gravity calculation, and the task-space Jacobian matches a finite-difference check — a rigid-body model that is subtly wrong in an axis convention or a sign is a classic, easy-to-miss failure mode, and the certificate's claims are only as good as the torque prediction underneath them. The same three-baseline structure as §VIII-A is used (B1 no handling, B2 reactive current-instant throttling, B3 the proposed architecture), sharing one computed-torque tracking controller so that the B2-vs-B3 comparison stays fair.

B. The certificate and hierarchy behave as designed

On a benign trajectory where the certificate never triggers, B1/B2/B3 are bit-for-bit identical (0.3mm final error, no saturation) — adding the predictive layer costs nothing when it is not needed. On the flagship scenario (§VIII-F's design: two routes to nearby targets under a fixed heavy payload, one through a persistently high-static-torque "outstretched" pose, one that stays in a low-demand "tucked" region), B1 and B2 attempt the shorter route and never reach the shared goal (peak torque ratio 1.00, final position error 0.94m and 0.82m, 45 and 54 saturation samples respectively); B3's certificate predicts before execution that $m_1^\star(P_A) < m_{\text{safe}}$ — retiming alone cannot restore margin along P_A, persisting even at 8x the maximum retiming factor tested — reroutes to the safe alternative, and reaches the identical goal with 0.001m final error and zero saturation samples. This is Theorem 3 exercised end to end on real (if reduced-order) dynamics: the pre-execution decision that triggers Level 3 here is exactly the bisection search over $\lambda \in \Lambda$ the theorem formalizes, not a placeholder.

The A1–A5 ablations (§VIII-H) separate out why. On a scenario where retiming alone suffices, A3 (predict without acting) is bit-for-bit identical to A1 (no prediction at all) — same saturation count, same failure — a clean confirmation that a certificate which only detects and never feeds back into the planner has zero effect on outcomes; whatever value prediction-as-monitoring might have (the diagnostic signals of §IV, not evaluated here) is a separate claim from the one this architecture makes. On the flagship scenario, A4 (predict, retime, and reshape, but Level 3/4 disabled) fails, and in fact fails *worse than doing nothing* (53 vs. 45 saturation samples, final position error 1.35m vs. A1's 0.94m, neither reaching the shared goal), while A5 (the full hierarchy) succeeds cleanly, reaching the identical goal with 0.001m error and zero saturation samples — isolating that rerouting specifically, not adaptation in general, is load-bearing here, exactly the separation §VIII-H's ablation design was built to produce. (A4's numbers moved from an earlier-reported 0.98m/47 once the Level-2 reshape QP's objective was corrected to penalize deviation from the nominal trajectory rather than raw acceleration magnitude — a code-vs-paper review found the original objective had no reason to prefer a reshaped motion close to the one the planner originally intended; see the repository README for the before/after comparison. The qualitative conclusion is unchanged and, if anything, stronger: reshaping without rerouting does not merely fail to help on this scenario, it actively makes the outcome worse.)

C. Detection lead time and conservatism

Under an unanticipated external disturbance (Exp. 3, §VIII-D), B2's reactive check fires only once torque is already at the limit ($T_{\text{warning}} = 0$ by construction); B3's certificate fires 0.42s earlier, using only a 0.3s bounded online prediction horizon, not oracle knowledge of the disturbance schedule. When the same class of disturbance is instead known in advance as part of the predicted environment (Exp. 4, §VIII-E: a contact-stiffness transition), B3 adapts before the transition occurs and finishes with zero saturation samples against 27 (B1) and 8 (B2) for the two baselines. Conservatism (§VIII-I): across a payload sweep spanning no-adaptation-needed to severe deficits (Exp. 2, §VIII-C), the certificate triggered adaptation at 13 of the tested payload levels, and zero of those 13 triggers were false positives. This is a single-scenario-family result, not a general conservatism bound, but it is the direction the architecture needs to succeed in, and is reported precisely because an overly conservative certificate could otherwise make every other number above look good while triggering fallback constantly. What this sweep does not measure is the false-negative side (§VIII-I): whether the certificate ever fails to trigger before a real violation occurs. No false negatives were observed incidentally across the experiments reported here, but none of them were designed to stress-test this specifically (e.g. via a deliberately understated $\Delta\tau_i$ or an environment-prediction error exceeding what was assumed), so this should not be read as a measured false-negative rate.

D. Negative and boundary results

Two findings argue against overstating the architecture's benefit and are kept here rather than dropped. First, **partial adaptation is not always better than none**: in the payload sweep, once payload exceeds the point where Level-1 retiming can fully restore margin (2.6–3.0kg in this platform), the partially-retimed trajectory leaves the arm in a *worse* final tracking state than the unmodified baseline (e.g. 189.6mm vs. 117.2mm final error at 3.0kg) — a certificate-triggered correction that only partially succeeds is not guaranteed to dominate doing nothing. Second, **Level 4 is a safety outcome, not a task-completion outcome**, and this is a real trade-off rather than an incidental implementation detail: consistent with Level 4's "terminal safe-set policy" framing (§V-B) and Theorem 4's deferred status, once triggered it holds the robot at its stopped configuration rather than resuming automatically — confirmed necessary during implementation, where an earlier version that let the certificate re-evaluate the original plan after a stop produced an unstable closed loop once the robot had genuinely diverged from it. The consequence: from 25N on in Exp. 3 (B1/B2/B3 all succeed through 24N; re-verified directly, and confirmed independent of the route-level-reshape work of §IX-H — identical with Level 2 disabled entirely), B3 fails the task via a permanent hold at disturbance magnitudes B1/B2 still complete despite transient saturation — a genuine safety-vs-completion trade-off, not a strict win. A further, physically real limitation surfaced in the same experiment: once a growing disturbance reaches its full magnitude, the *static* holding torque at whatever pose Level 4 stopped in can itself exceed the actuator limit — a fixed hold provides no guaranteed robustness against a disturbance that outgrows what can be held statically, and closing this would need either a Level 4 that re-picks its holding pose or a coupling to Level 3 rather than freezing at the first trigger, left as future work. Third, at the payload sweep's actual task-success crossover (2.4kg — corrected after rerunning at finer resolution than an earlier pass, which had mistakenly reported 3.0kg), B2's reactive throttle succeeds exactly as well as B3's predictive retiming; this specific scenario does not demonstrate an advantage for prediction over reaction, which instead shows up in Exp. 3–5, not uniformly.

E. Real-time cost of the certificate

On the benign trajectory, B3's online per-cycle cost is negligible (mean 0.27ms against the 20ms/50Hz budget used throughout §VIII). Under the flagship stress case, the online step that matters is a single Level-2 QP solve triggered before the certificate falls back to (sticky) braking: pooled over repeated measurements, this one-shot solve costs roughly 27–54ms — 135–270% of the budget, i.e. it EXCEEDS the budget outright rather than merely approaching it — and disabling Level 2 isolates that the QP (solved via a general-purpose convex solver that reconstructs the optimization problem from scratch on every call rather than reusing a compiled, parametrized form) accounts for approximately 100% of the online-step cost whenever it fires. (This cost roughly doubled after a code-vs-paper review found the QP was not dynamically consistent – see §V-B's Level 2 description – and fixing it added state-trajectory variables and integration constraints to the same problem; the increase is the honest price of the QP now certifying a trajectory it can actually produce, not a regression to be hidden.) This is reported as a genuine, unresolved implementation limitation: a single Level-2 solve costs more than the entire real-time budget on its own, and closing this gap would need a parametrized/warm-started QP formulation before any real-time deployment against §VII's MoveIt integration.

F. What this does and does not establish

This is a 3-DOF planar reduced-order platform built to check that the certificate, hierarchy, and Theorem 3's rerouting claim behave as designed on a system simple enough to verify independently at each step — not the FR3-scale evaluation §VIII specifies. §VIII's own benchmark design, ablation set, and real-time protocol were followed as written for this reduced-order system. What this platform does not establish: results at real FR3 scale (§IX-I now covers Experiments 1-3 of that gap, on the real MoveIt 2 integration of §VII, but not 4-7 or the real-time/ablation protocol at that scale), the obstacle/collision feasibility set $\mathcal{F}_{\text{obs}}$ (out of scope for this reduced-order study, which addresses $\mathcal{F}_{\text{dyn}}$ only, and also out of scope for §IX-I), and any legged/humanoid extension. Complete code, every scenario parameter behind the numbers above, and a documented account of three implementation bugs found and fixed during this verification effort — kept visible rather than silently corrected, in keeping with this draft's own standard for what stays open — are available in the accompanying repository.

G. Environment-conditioned rerouting

Added in a follow-up pass after a code review identified a genuine gap: §IX-B's flagship result and §VIII-F's design both reroute because of a configuration/payload deficit, and §VIII-E's known-transition experiment shows adaptation but only ever gives the planner one route — neither shows rerouting *because of the environment specifically*. Experiment 7 (§VIII-L) closes this: P_A and P_B share q_0 and q_g as in the flagship scenario, but here P_A's via-point has a healthy static margin under the (light, 1.0 kg) payload alone (**9.87 Nm**) — unlike the flagship scenario, this is not a configuration/payload deficit — and becomes infeasible (**-20.22 Nm**) only once the same scripted contact-stiffness force from §VIII-E is included as part of the known predicted environment; P_B's via-point never enters the field and its margin is unaffected (**8.63 Nm**, force or no force). The certificate's own bounded retiming search confirms the deficit is retiming-proof, for the same reason as the flagship scenario's static deficit: the force is a function of end-effector position, evaluated at a zero-velocity/zero-acceleration via-point boundary, so it does not shrink under any retiming factor. B1 and B2 both attempt P_A, saturate through the field (peak torque ratio 1.00, 40 and 44 saturation samples), and fail to reach the goal (final position error 0.234m and 0.366m); B3, given the same contact model at planning time as §VIII-E's experiment (`force_known_at_plan_time=True`), predicts the deficit before ever entering the field, reroutes to P_B, and reaches the identical goal with 0.000m final error and zero saturation samples, `replans=1`. This is the same Theorem 3 mechanism as the flagship result, exercised on a genuinely different cause: an environmental feature known in advance, not the manipulator's own kinematics or payload.

H. Route-level reshape before rerouting

Added after a conceptual discussion, not a code review, identified a genuine architectural gap: the pre-execution route decision (`plan_route`) previously tried only retiming before escalating to reroute, even though reshaping (Level 2) is, like retiming, a form of regenerating the SAME route rather than changing it, and was only ever tried online, per control cycle, never at route-planning time — the gap §V-C already flagged ("closing that expressiveness gap ... is identified as a concrete next step, not yet implemented"). `plan_route`'s order is now retime, then reshape, then reroute, on the primary route and (if reached) the alternate, symmetrically. The reshape check is a whole-route version of the same QP used online, extended with a terminal-state constraint pinning it to the SAME goal at rest — without which a whole-route reshape could silently drift to a different final configuration, reopening for Level 2 the exact goal-changing failure mode already fixed once for Level 3 (§IX-B/Draft Status item 9).

Three findings, each checked directly rather than assumed. First, reshape genuinely cannot resolve either of this benchmark's existing reroute-required scenarios: on both the flagship scenario (§IX-B) and Experiment 7 (§IX-G), the whole-route reshape QP fails to restore $m_{\text{phys}} \geq m_{\text{safe}}$ — confirmed against SCS as an independent solver, not merely OSQP's own status flag, since OSQP alone reports an inconclusive iteration-limit status on a problem this size rather than a clean infeasibility certificate. Level 3 rerouting therefore remains genuinely necessary in both cases, not an artifact of reshape never having been tried. Second, a scenario exists where reshape succeeds and retiming does not: a route entering the same scripted contact-force field as Experiment 7, but with a lighter force constant and a more generous route duration, is retiming-proof (the deficit does not shrink under any retiming factor, as in Experiment 7) yet reshape restores $m_{\text{phys}} = 2.55 \geq m_{\text{safe}}$ while reaching the identical goal. Notably, it does not do this by geometrically avoiding the field — the reshaped path's minimum end-effector height is, if anything, slightly lower than the nominal path's — but by reallocating velocity and timing within the fixed total route duration, a mechanism genuinely distinct from retiming's uniform global time-scaling.

Third, and the most consequential for deployment: this whole-route QP is materially larger than the online horizon's (its size scales with route duration divided by the discretization step, not a fixed online horizon length), and OSQP does not reliably converge on it within a practical iteration budget — confirmed directly, hitting an iteration-limit status even at 50,000 iterations on the flagship scenario, while SCS solves the identical problem to a clean optimum in under a second. With an OSQP-then-SCS fallback, the one-time, pre-execution cost of the route decision rises from retiming alone (roughly 1–3ms when the fast bisection path applies, up to roughly 80ms on scenarios where the dense-grid non-monotonicity fallback of Draft Status item 14 is needed) to the range of 300ms–900ms once reshape is attempted, paid whether reshape ultimately succeeds or fails. This remains a one-time cost, not a per-cycle one, and so is not compared against the 20ms real-time budget elsewhere in this section — but at FR3-scale route durations it could plausibly become the dominant term in how long a robot waits before moving, which is disclosed here rather than left implicit. This finding, together with the second, is exactly the shape of evidence the conceptual discussion that motivated this addition anticipated: predicted authority loss should not be assumed to require rerouting, but assessing whether regeneration suffices is not free, and that cost is now measured rather than assumed away.

I. FR3-Scale MoveIt 2 Hybrid Planning Platform: Real Results for Experiments 1-3

A second, independent verification line was built after §IX-A–H: not a reduced-order proxy, but a real integration of §VII's target architecture — C++ `LocalConstraintSolverInterface`/`TrajectoryOperatorInterface` plugins for B2 and B3, a real Franka FR3 URDF/kinematic/dynamic model (shared inverse-dynamics code path between B2 and B3, so the comparison stays fair as in §IX-A), MoveIt 2 Hybrid Planning with OMPL as the unmodified global planner, and MuJoCo-simulated dynamics for execution. This platform currently exercises Levels 0/1/2/4 (pass-through, retime, reshape, sticky brake) and the certificate itself; Level 3 (reroute) has not been exercised here, only on the reduced-order platform (§IX-B, §IX-G). Complete code, build instructions, and a full account of every bug found and fixed while building it are in the accompanying repository (`ros2_ws/`, gitignored from this paper's own repository since it is engineering infrastructure rather than paper content, with its own git history).

Methodology note shared by all results below. An early version of Experiments 2 and 3 on this platform triggered a fresh, unseeded OMPL plan per cell/baseline, which does not guarantee the "identical geometric trajectory, geometric path and time law held fixed" comparison both experiments' own designs (§VIII-C, §VIII-D) require. This was found and fixed: one trajectory is now planned once via the stock, unmodified global planner and replayed verbatim (a dedicated `ReplayGlobalPlanner` global-planner plugin deserializes and re-publishes one captured `moveit_msgs/MotionPlanResponse`) for every subsequent cell in a sweep. Verified deterministic directly, not assumed: two independent fresh replay launches of the same captured trajectory produced byte-identical global-trajectory messages and final joint positions matching to ~3×10^{-7} rad. All numbers below use this replayed-trajectory methodology.

Experiment 1 (no regression). On the standard within-limits goal, B1 (stock MoveIt), B2, and B3 all succeed with near-identical final tracking error and zero online interventions for B2/B3 — the same "adding the predictive layer costs nothing when it is not needed" result §IX-B reports on the reduced-order platform, now confirmed at real FR3 scale with real MoveIt 2 machinery in the loop. (The real controller has a genuine steady-state joint-space tracking residual, $\approx$ 0.045-0.05 rad L2, stable across a settle window rather than transient; the success tolerance used throughout this platform is set from this measured floor rather than an idealized zero.)

Experiment 2 (payload sweep, activation-threshold separation). The identical replayed trajectory is re-executed at end-effector payloads $\{0, 3, 5, 6, 7, 8, 10\}$kg. B3's certificate first reports an online intervention at 6kg; B2's reactive check first intervenes at 10kg — a 4kg separation in activation threshold (m_{phys} trending $9.65 \to 5.79 \to 3.21 \to 1.91 \to 1.10 \to -0.10 \to -2.54$ N·m over the sweep, reproducing to the stated precision across independent reruns). This is reported as an activation-threshold separation rather than a "prediction lead time," deliberately: B2 and B3 trigger on different conditions by construction (B2 on current-instant torque already exceeding the limit with no margin, B3 on a predicted future margin falling below m_{safe}), so part of the 4kg gap reflects that threshold difference, not solely B3's predictive horizon — the genuine lookahead claim is reserved for Experiment 3 below, where both baselines share the same triggering logic (current-instant torque) and differ only in whether the disturbance is known in advance. A caveat on task completion specific to this platform: the payload-sweep goal used here is large enough in joint-space travel to encounter a platform-level execution-fragility limitation discussed under Experiment 4 below, so this experiment measures the payload at which each baseline's own trigger condition first fires within a fixed per-cell time budget, not full task completion at every payload — consistent with the primary metric §VIII-C specifies (first violation/first trigger), not a stronger claim.

Experiment 3 (interaction force, genuine detection lead). Under an unanticipated ramp disturbance applied during execution of a within-limits goal, B3 detects the coming violation $\approx$ 0.32-0.34s after local execution begins; because the disturbance itself starts partway into that window (not at execution onset), this is $\approx$ 0.06s after the disturbance itself begins, giving $T_{\text{warning}} \approx$ 0.40-0.41s. B2 never detects the violation before it occurs. (An earlier internal labeling of this number as time "after force onset" was incorrect and is corrected here — it measures time after local execution begins, per the platform's own repository history.) Beyond the detection-lead comparison itself, this platform adds a check §IX's reduced-order results do not report: a direct predicted-vs-observed accuracy validation for the certificate's own future-margin prediction, comparing m_{phys} (predicted, at plan time) against m_{phys} re-evaluated at the later real measured state once that future cycle actually arrives. Over the validated predictions in this run (a small sample, $n = 4$, from one run's own online window — reported as what it is, not extrapolated), RMSE $\approx$ 0.19 N·m, MAE $\approx$ 0.18 N·m, mean error $\approx$+0.10 N·m, max absolute error $\approx$ 0.24 N·m, at a 0.28s prediction horizon. This validates prediction accuracy under measured robot-state evolution specifically — the force input to both the prediction and its later comparison point is the same commanded/modeled force, not an independently measured one — not a complete independent validation of the disturbance model itself; that distinction is stated here rather than left for a reader to over-read from the RMSE alone.

Experiment 4 (known contact transition): blocked, and why. This experiment's implementation is complete but not currently runnable on this platform, for a reason unrelated to the certificate architecture: goals with enough joint-space travel to exercise a meaningful end-effector transition run into a platform-level actuator-effort saturation limitation, root-caused directly rather than left as an unexplained failure. The proximate mechanism is a sustained relay/bang-bang oscillation — every joint's commanded PID effort sits at or beyond its real torque limit 42-60% of run time, peaking at 2.7-3× over budget — traced to the stock controller's gains, copied from the real robot's own configuration, demanding more corrective torque per unit tracking error than the low-torque wrist joints' budget (12 N·m) can supply once meaningful tracking error has built up. A gravity-and-Coriolis feedforward controller built specifically to address this eliminates the oscillation outright (extrema/sec falling from 22.5-57.1 to 0-0.45) and is a genuine, permanent improvement over the stock controller, but does not by itself achieve full task completion on the largest goals tested; the remaining limitation was isolated directly in MuJoCo to a real, sharp per-joint torque threshold rather than a tuning gap (see the repository's own root-cause account). One finding from this investigation is worth surfacing here independent of whether the goal itself is fixed: **the certificate has a real, disclosed blind spot with respect to this specific failure mode.** m_{phys} (and its state-conditioned counterpart, m_{phys} re-evaluated at the real measured state) both evaluate the torque the *idealized reference trajectory* requires, not what the real closed-loop tracking controller ends up commanding once tracking error and controller gain interact — during the oscillation described above, the certificate reported a healthy margin (9.65-10.0 N·m, never below 4.0) for the entire duration of a run in which the real commanded effort was saturating on every joint. This is not a bug in the certificate as specified (§IV conditions m_{phys} on predicted state, contact, payload, and disturbance — a controller-induced saturation pathology is none of those), but it is a genuine scope boundary this platform surfaced that the reduced-order study's own computed-torque tracking controller (§IX-A) does not have occasion to expose, and is reported here as exactly that: a limitation of what the certificate as currently scoped can see, not a claim that it should have caught this.

Experiments 5-7 (flagship reroute, adaptation-continuum, environment-conditioned reroute): not attempted on this platform. These require Level 3, which this platform's `TrajectoryOperatorInterface` plugin implements as a mechanism (§VII) but has not yet been exercised in an FR3-scale scenario; §IX-B and §IX-G report these mechanisms working as designed on the reduced-order platform, which remains the only evidence for Theorem 3's rerouting claim and the environment-conditioning contribution (§IV) at any physical scale as of this draft.

Real-time cost. Unlike §IX-E's reduced-order measurements, per-cycle solve-time costs for the C++ plugins on this platform have not yet been systematically characterized and are not reported here; this is an open item, not a claim of negligible cost by omission.

X. Discussion

Conservatism versus responsiveness. Every mechanism in §V trades safety margin against unnecessary intervention; §VIII-I's conservatism metric is the paper's answer to the concern that any predictive-safety architecture can trivially "solve" every failure mode by refusing to act. §IX-C's reduced-order sweep reported zero false-positive triggers, a promising direction but, being a single-scenario-family, reduced-order result, not yet the general conservatism bound this concern needs answered at FR3 scale. This is also only half the picture: the metric bounds false positives, not false negatives (a real violation the certificate fails to catch in time), and no experiment in this draft was designed to stress that side specifically — reported here as an explicit gap, not implied to be zero by the absence of an observed counterexample.

Model and environment-prediction uncertainty. Theorem 1's guarantee is exactly as strong as the uncertainty bound $\Delta\tau_i$ used to construct the robust margin; if the environment prediction $E_{k:k+N}$ is itself wrong (e.g., an unanticipated contact), the certificate degrades gracefully to the accuracy of that prediction rather than failing catastrophically, but this claim is not yet backed by an experiment specifically stressing environment-prediction error and should not be asserted more strongly than that.

Sensing latency and computational scaling. §IX-E's reduced-order measurement found the online certificate check itself negligible but a single Level-2 QP solve costing close to the entire real-time budget on its own — a genuine, unresolved cost that should be treated as representative of the concern this paragraph raises, not fully addressed by it: the FR3-scale measurement called for in §VIII-J remains to be done, and a parametrized/warm-started QP formulation is the most likely fix if the same bottleneck reappears at scale.

Relationship to learned/world-model planners. §XII sketches a longer-term architecture in which a learned high-level planner (a world model or vision-language-action policy) determines *what* the robot should do, this architecture's local planner determines *how* it should move, and the predictive-realizability layer determines *whether* the robot can physically execute that behavior under current and predicted conditions. This is presented as a long-term direction, not a near-term deliverable of the present paper, and no learned-planner integration has been attempted.

XI. What This Paper Does Not Claim

To keep the contribution claims defensible, this paper explicitly does not claim: to be the first torque-aware or actuator-aware motion planner; that MoveIt cannot handle dynamic constraints; that the proposed method guarantees safety under arbitrary disturbance; that predictive-saturation feedback guarantees a legged platform will not fall; that no prior work has considered torque limits in planning; or that this architecture is suitable for tasks whose optimal behavior operates at or near the actuator limit by design — minimum-time trajectories and other margin-spending (rather than margin-preserving) tasks are out of scope (§III), since the certificate as defined here cannot distinguish an intentionally optimal boundary solution from an unintended loss of control authority, and would drive the former away from its own optimum. Reference and Explicit Reference Governors have predicted a margin and fed it back before constraint violation since the 1990s (§II-C); retiming under predicted torque saturation is an established sub-field (§II-D); and terrain/contact-conditioned actuation limits in legged planning have prior published treatment that this paper must cite and differentiate from, not merely acknowledge in passing (§II-D). The claim this paper makes is narrower: a predictive feedback interface that converts future physical-realizability information from the execution layer into motion-planning adaptation and, when necessary, route-level replanning, unified across geometric and physical failure modes under one hierarchical response — not the existence of torque limits, margins, or saturation constraints considered individually.

XII. Long-Term Direction (not a deliverable of this paper)

$$\text{World model / VLM} \rightarrow \text{Global motion planner} \rightarrow \text{Predictive physical realizability}$$
$$\rightarrow \text{Whole-body / interaction controller} \rightarrow \text{Robot + environment},$$

with the loop closed by re-planning on the realizability layer's own feedback. The high-level system decides *what* the robot should do; the planner decides *how* it should move; the realizability layer decides *whether* the robot can execute that behavior under current and predicted conditions. This is proposed as a long-term organizing idea for a model-based Physical AI stack, not a contribution defended by any result in this draft.

XIII. Conclusion

Motion planning pipelines certify that a trajectory can be geometrically generated and kinematically executed; they do not, in general, continuously ask whether the robot will remain physically capable of realizing that trajectory as configuration, contact, payload, and disturbance evolve. This paper proposes predictive physical realizability as the missing feedback signal — a receding-horizon certificate on actuator margin, conditioned on predicted environment and contact information, coupled to a hierarchical planner response that adapts before rerouting and reroutes before falling back, with an explicit theorem-backed condition for when adaptation is provably insufficient. The architecture is designed to occupy an existing, currently-empty slot in MoveIt 2's own Hybrid Planning interface rather than propose a new integration target. What remains is empirical: whether the certificate's prediction lead time is large enough to matter in practice, and whether the conservatism cost of acting on it is small enough to be worth paying — both questions §VIII's benchmark design is built to answer, and both given a first, reduced-order answer in §IX-A–H (a 0.42s detection lead on an unanticipated disturbance; zero false-positive triggers across a 13-point sweep) that is encouraging but explicitly not a substitute for the FR3-scale study §VIII specifies. A second, partial answer now also exists at real FR3 scale (§IX-I): a real MoveIt 2 Hybrid Planning integration reproduces the no-regression result (Experiment 1), an activation-threshold separation under payload variation (Experiment 2), and a genuine ≈ 0.40s detection lead together with a direct predicted-vs-observed accuracy check on the certificate's own future-margin prediction (Experiment 3) — while also surfacing, honestly, a real scope boundary the reduced-order platform's simpler tracking controller never had occasion to expose: the certificate cannot see torque demanded by closed-loop controller error itself, only by the idealized reference trajectory, which blocked Experiment 4 on this platform for reasons unrelated to the certificate's own correctness. §IX-A–H also surfaced the trade-offs a reduced-order platform makes visible early and cheaply: partial adaptation is not always better than none, a terminal safety response trades task completion for safety rather than guaranteeing both, and a single QP solve in the current implementation costs close to the entire real-time budget on its own. None of these close the gap to a submission-ready result — Level 3/rerouting and Experiments 4-7 remain unvalidated at FR3 scale, and real-time cost has not yet been characterized on that platform — they are the reason the honest next step is finishing the FR3-scale study, not a claim that this draft has already completed it.

References

[1] M. Prats, "TOTG not respecting acceleration limits," moveit/moveit2 GitHub Issue #2600, reported Dec. 2023, closed Jan. 2024. [Online]. Available: https://github.com/moveit/moveit2/issues/2600

[2] E. Garone, S. Di Cairano, and I. Kolmanovsky, "Reference and command governors for systems with constraints: A survey of their theory and applications," *Automatica*, vol. 75, pp. 306–328, 2017.

[3] A. Bemporad, A. Casavola, and E. Mosca, "Nonlinear control of constrained linear systems via predictive reference management," *IEEE Transactions on Automatic Control*, vol. 42, no. 3, pp. 340–349, 1997.

[4] S. Zhang, J. Y. Z. Liu, and D. Liao-McPherson, "Integrating Planning and Predictive Control Using the Path Feasibility Governor," arXiv:2507.09134, Jul. 2025. Submitted to *IEEE Transactions on Automatic Control*.

[5] R. Orsolino, M. Focchi, S. Caron, G. Raiola, V. Barasuol, and C. Semini, "Feasible Region: an Actuation-Aware Extension of the Support Region," *IEEE Transactions on Robotics*, vol. 36, no. 4, pp. 1239–1255, Aug. 2020. arXiv:1903.07999.

[6] B. Acosta and M. Posa, "Perceptive Mixed-Integer Footstep Control for Underactuated Bipedal Walking on Rough Terrain," *IEEE Transactions on Robotics*, vol. 41, pp. 4518–4537, 2025. arXiv:2501.19391.

All six entries above were independently verified against their primary source (GitHub issue pulled directly via `gh issue view`; papers checked via arXiv/publisher pages) during this draft's review process, not copied from the source planning documents without checking — see *Draft Status and Open Items*, items 1 and 4, for what that verification did and did not confirm. Two further claims in §II-D (disturbance-observer-based retiming "in the early 2000s"; a task-space torque-limit projection for non-flat-terrain trajectory optimization) are carried over from the source planning document without a specific citation and are not included here — see *Draft Status and Open Items*, item 8.

Draft Status and Open Items

This section tracks what remains open so it cannot be silently dropped in a later editing pass. Items now fully resolved and already reflected in the main text are noted with a pointer rather than re-narrated in full; only genuinely open content is described at length. (All numbers cited anywhere in this draft were re-verified against a fresh run of the accompanying code, `code/run_all.py`, immediately before this pass — every headline figure in §IX-A–H matches exactly, and `pytest tests/` passes 14/14.)

1. **Citations.** Only moveit/moveit2#2600 ([1]) is a confirmed, independently-pulled data point (§I, §II-A); two other MoveIt 2 issues referenced in an earlier pass could not be re-located and are not cited. Both places that risked overclaiming what #2600 shows — §I and §II-A — now state the edge-case/`path_tolerance` caveat inline rather than presenting it as general evidence.
2. **The Level 3 mechanism gap (§V-C) is partially closed.** Resolved: the original Theorem 3 was near-tautological and has been replaced with a code-verified, non-trivial sufficient condition (§VI). Still open, per §V-C: what the rerouter should do once triggered (no proof that geometric-escape targets restore $\mathcal{F}_{\text{dyn}}$, no explicit physical-feasibility objective on route selection), and folding Level 2 into the formal adaptation class $\mathcal{A}(p)$.
3. **Theorem 4 (recursive feasibility) is deferred** — not proved, not disproved, not attempted. Whether to attempt it, or present Level 4 as an empirically-validated rather than formally certified fallback, is undecided.
4. **Prior art — PathFG, Orsolino/Focchi, and Acosta/Posa have all been read in full**, not summarized from an abstract, and §II-C/D's differentiation from each reflects that direct reading. One unverified detail remains: [4]'s "submitted to *IEEE Transactions on Automatic Control*" line is from arXiv listing metadata, not the paper's own text, and should be re-checked against the arXiv abstract page before submission.

5. **Legged/humanoid resourcing is unresolved.** No legged/humanoid pipeline exists in this line of work; §VIII-K scopes all designed experiments to the fixed-base manipulator until a real build-cost estimate is obtained.
6. **Publication sequencing.** Per the source planning document, this paper should not proceed to substantive external submission until `mp_main` (ICRA 2027) and HAE (IJSS) have received decisions; the recommended next concrete step is a separate, narrower saturation-certificate paper proving Theorems 1–3 independent of this paper's broader scope.
7. **§VIII's FR3-scale experiments are only partially run** — see §IX's own introduction and §IX-F/§IX-I, not restated here. In short: the reduced-order platform (§IX-A–H) covers the full design at reduced scale; the real FR3-scale MoveIt 2 platform (§IX-I) covers Experiments 1–3 at real scale. Level 3/rerouting and Experiments 4–7 remain validated only at reduced scale.
8. **Two claims in §II-D still lack a specific citation**: disturbance-observer-based retiming "in the early 2000s," and a task-space torque-limit projection for non-flat-terrain trajectory optimization (distinct from the now-cited Orsolino/Focchi work). A specific paper must be identified for each before submission, or the claims removed/rephrased as general background.
9. **Two implementation bugs found by external code review, now fixed, with §IX's numbers updated to match.** (a) Experiments 5/6's "reroute" was changing the goal, not the route — P_A/P_B ended at different final configurations, so B3 could pass by abandoning the task for an easier target. Fixed with a two-segment via-point representation so both genuinely share q_0/q_g (the same bug, also present in the ablation-batch code, was fixed there too). (b) The replanning-count metric counted a persistent flag on every control cycle rather than discrete events; fixed to count the route decision, brake engagement, and Level-2 corrections as distinct, rising-edge events. Separately: Level 2's reshape QP previously optimized accelerations against the nominal trajectory's $(Q, \dot{Q})$ held fixed, with no constraint tying the result back to a trajectory it could actually produce; fixed by adding state variables and integration constraints to the same QP — now reflected directly in §V-B's formula, and the roughly-doubled Level-2 timing figures in §IX-E are the honest cost of that fix. B2's one-step throttle was also found not to guarantee its own claimed torque bound (the gravity/Coriolis/external-force term does not scale with the correction) and was replaced with an exact one-step QP projection.
10. **A follow-up review, run from scratch against the fixed codebase, found the fixes sound and four smaller issues, now addressed**: the Level-2 objective correction is item 9's own fix, now in §V-B directly, not restated here. The remaining three are code-hygiene fixes with no numeric consequence for any claim in this draft — a code comment that could have been read as overclaiming Theorem 3's idealized case rather than its actual sufficient condition (tightened), `saturation_events` renamed to `saturation_samples` (it counts samples, not episodes), and `min_actuator_margin_nm` split into commanded/applied variants so the name matches what each measures.
11. **Experiment 7** (§VIII-L, results in §IX-G) was added to close a gap the same review identified: Experiments 5/6 both reroute because of a configuration/payload deficit, and Experiment 4 shows adaptation but only ever on a single route — nothing showed rerouting *because of* the environment specifically. §IX-G reports the result; not restated here.
12. **Route-level reshape before rerouting** (§V-B/§V-C, results in §IX-H) was added after a conceptual discussion concluded that trajectory regeneration on the current route should be tried before escalating to reroute, and the pre-execution route decision previously never tried reshaping. §IX-H reports the three findings (reshape cannot resolve either existing reroute scenario; a separate scenario exists where reshape succeeds and retiming does not; the whole-route QP costs 300–900ms, one to two orders of magnitude above retiming alone); not restated here.
13. **One genuine numeric imprecision was found and corrected**: Experiment 3's safety-vs-completion boundary was originally reported off by 1N at the low end and ~5N at the high end; §IX-D's "B1/B2/B3 all succeed through 24N" already reflects the corrected value. A subsequent full code-and-paper review otherwise confirmed the draft's numbers against fresh reruns and re-read `dynamics.py`, `certificate.py`, `baselines.py`, and `executor.py` in full with no further issues found.
14. **A real gap in Theorem 3's monotonicity assumption was found, verified against the implementation, and fixed** — this is now the Lemma in §VI, which states the precise per-joint sign condition and the ~0.8% failure rate found in a 3000-scenario random search. Not restated here. Neither result actually cited in this paper (the flagship scenario, Experiment 7) changed: both

were checked directly against the Lemma's condition. The exact closed-form fallback sketched in `monotonicity_lemma_draft.md` (evaluating $\{1, \lambda_{\max}\} \cup \{\lambda_i^\star\}$ directly rather than a dense scan) is being developed for the spin-off paper of item 6 and is not what §VI's dense-scan fallback claims.

15. **Two scope statements were added and are now in the main text**: the margin-as-asset non-claim (§III, §XI) and the explicit scoping of the fixed-structure computational claim to certificate evaluation and Level 0–1 only (Abstract, §VI). Not restated here.
16. **A cross-paper margin-stacking question is flagged, not resolved.** The source planning document's long-term stack (§XII) places this paper's m_{safe} margin in series with an execution-layer margin from a separate, concurrently-developed paper (the pHRI/interaction-dynamics line, not part of this repository). Whether the two margins' scope is already disjoint or needs explicit reconciliation has not been checked; that draft must be read before any joint submission or cross-reference.
17. **The conservatism metric's false-negative gap is now stated in the main text** (§VIII-I, §X): the metric bounds false positives only; no experiment here was designed to stress-test false negatives specifically. Not restated here.